Photodiagnosis and Photodynamic Therapy 17 (2017) 87–91

Contents lists available at ScienceDirect

Photodiagnosis and Photodynamic Therapy

journal homepage: www.elsevier.com/locate/pdpdt

Review

Photodynamic therapy as an alternative treatment for relapsed or refractory mycosis fungoides: A systemic review

Jingwen Xue[a], Cynthia Liu[b], Yuehua Liu[c,*]

[a] School of Medicine, Tsinghua University, Beijing, 100084, China
[b] University of Missouri - Kansas City School of Medicine, Kansas City, MO, United States
[c] Department of Dermatology, Peking Union Medical College Hospital, Peking Union Medical College and Chinese Academy of Medical Sciences, Beijing 100730, China

ARTICLE INFO

Article history:
Received 11 September 2016
Received in revised form 18 October 2016
Accepted 22 November 2016
Available online 23 November 2016

Keywords:
Photodynamic therapy
Relapsed or refractory mycosis fungoides
Aminolevulinic acid
Methyl aminolevulinic acid

ABSTRACT

Mycosis fungoides is the most common cutaneous T-cell lymphoma. It is characterized by slow progress over years to decades, developing from patches to infiltrated plaques, and sometimes to tumors. Therapies such as localized chemotherapy, photochemotherapy and radiotherapy are often employed when lesions of refractory or relapsing mycosis fungoides are resistant to conventional therapies. However, these methods have acute or chronic side effects and toxicity, which may accumulate with repeated and protracted treatment cycles. Photodynamic therapy is a promising, well-tolerated option for the treatment of localized lesions with excellent cosmetic outcomes. In this article, we systematically reviewed and discussed clinical application of photodynamic therapy in relapsed or refractory mycosis fungoides. There are 20 papers included in this review article.

© 2016 Elsevier B.V. All rights reserved.

Contents

1. Introduction 87
2. Method 88
   2.1. ALA-photodynamic therapy 88
   2.2. MAL-photodynamic therapy 88
   2.3. Light sources 88
3. Result 90
4. Discussion 90
   Funding support 90
   Conflict of interest disclosure 90
   Acknowledgements 90
   References 90

1. Introduction

Mycosis fungoides (MF) is a relatively rare, extranodal, non-Hodgkin's lymphoma with primary cutaneous involvement, which is generally indolent in behavior. It is a low-grade cutaneous T-cell lymphoma (CTCL) characterized by small to medium sized cells with cerebriform nuclei that usually express mature peripheral CD4+ T-helper cell immunophenotypes and infiltrate the skin characteristically with epidermotropism [1]. The incidence rate of MF has been reported to be 4.1/1, 000, 000 by a SEER (Surveillance, Epidemiology and End Results) based study which used data from 16 registries in the United States to analyze 3884 cases of CTCL from 2001 to 2005 [2]. This makes MF the most common cutaneous T-cell lymphoma (CTCL) subtype (54%). The risk of MF is associated with age, gender and race, with relatively higher prevalence among the elderly, blacks and males [3]. Occupational exposure, genetic disposition, radiation, and viral infection have been speculated as

* Corresponding author at: Department of Dermatology, Peking Union Medical College Hospital, Peking Union Medical College and Chinese Academy of Medical Sciences, 1 Shuaifuyuan, Wangfujing, Beijing 100730, China.
E-mail address: yuehualiu@263.net (Y. Liu).

http://dx.doi.org/10.1016/j.pdpdt.2016.11.010
1572-1000/© 2016 Elsevier B.V. All rights reserved.

etiologic factors in relation to MF, while corresponding pathological mechanism is yet to be elucidated [4].

The appropriate treatment of MF is primarily based on prognostic factors, Tumor-Node-visceral Metastases-Blood (TNMB) classification, and overall clinical stage at diagnosis [5]. The major prognostic factors reported by a large cohort of 1502 patents included increased age, male sex, increased lactate dehydrogenase (LDH), large-cell transformation and folliculotropic MF. These factors all reduce survival and increase the risk of disease progression (RDP). Positive prognostic factors include hypopigmented MF, MF with lymphomatoid papulosis, and poikilodermatous MF. These show improved survival and reduced RDP [6]. The latest version of staging and classification was updated on 2008 by the World Health Organization (4th edn) and revised in 2010 [7,8].

MF constitutes the protracted clinical evolution of patches, plaques, and tumors with a wide spectrum of potential clinical manifestations. MF patients are usually treated with a multimodal approach. However, many patients go on to develop resistance against at least one type of treatment or have relapsing symptoms after a period of improvement. The treatment of relapsed or refractory MF has been very challenging due to the limited novel therapies available. In the past decade, clinical success has been reported using photodynamic therapy (PDT) as a possible alternative non-invasive, target-specific therapy to treat MF. PDT utilizes reactive oxygen species generated by light-activated photosensitizer to oxidize essential cellular components and thereby causes apoptosis and necrosis restricted to target sites.

In 1994, Svanberg et al. reported first successful case of using topical δ-amino levulinic acid (ALA)-based PDT to treat an MF plaque [9]. In 1994, Boehncke et al. found that the capability of PDT to inhibit proliferation of T cells was similar to that of PUVA both in *vitro* and in *vivo* [10]. In 1995, Rittenhouse-Diakun et al. demonstrated that malignant CTCL cells with increased CD71 expression, a marker of relative intracellular iron deficiency, preferentially accumulated endogenously generated protoporphyrin IX (PpIX) and were highly susceptible to ALA-PDT [11]. In 2001, Gad et al. showed an increase in caspase-3-like activities and increase in the percentage of DNA fragmentation in malignant T cells following ALA-PDT, which indicated that ALA-PDT could induce apoptosis and caspase activation [12]. Orenstein et al. observed different patterns of PpIX fluorescence in patients with early and advanced stages of the disease using 5-aminolevulinic acid topical application: in stage I MF lesions, no PpIX fluorescence was detected one hour after photoirradiation (580–720 nm), while in thick stage III lesions, PpIX fluorescence disappeared after an additional 10–15 min irradiation [13]. All these studies suggest the ability of PDT to selectively destroy the malignant lymphocytes in MF lesions.

2. Method

We performed a systematic review of literature on May 25, 2016 and updated the search on September 10, 2016. We searched keywords “mycosis fungoides” and “photodynamic therapy” in PubMed (National Library of Medicine), Google-Scholar, Scopus, and Web of Knowledge databases. An article was considered to be eligible for inclusion if it met the following criteria: (1) therapy-resistant mycosis fungoides; (2) clinical PDT application; (3) English language. Review articles were excluded. A total of 20 papers were included in this review.

2.1. ALA-photodynamic therapy

In 1990, Kennedy and colleagues discovered and reported the porphyrin precursor ALA. This finding was an important milestone in the development of PDT in dermatology, as the small molecules could easily penetrate the epidermis due to relatively low molecular weight [14]. ALA can be metabolized into photoactivatable porphyrins by most cells as part of the heme cycle, and the levels of intracellular accumulation are relatively higher inside altered cells constituting the diseased tissue [15].

The first case using ALA-PDT for successfully treating MF lesions was reported in 1994 [9], and since then many reports of successful ALA-PDT for patch/plaque stage MF have been published with good results. However, a number of limitations hinder the further application of ALA in PDT. For instance, because of poor tissue penetration, ALA cannot reach tumor tissue thicker than 3 mm in order to induce sufficient necrosis. Moreover, ALA requires an incubation period between application and light exposure to be metabolically converted into the active PpIX, which causes the patients to wait for a long time [16].

2.2. MAL-photodynamic therapy

A derivative of ALA, methy ester methyl aminolevulinate (MAL), is reported to have increased lipophilicity and deeper skin penetration when compared with ALA. This allows a shorter incubation time and results in a better selectivity toward tumor cells [17]. In 2013, Ariel et al. reported the first case of tumor-stage MF lesions that responded successfully to topical MAL-PDT treatment [18]. MAL-PDT is therefore considered as an alternative option in selected cases of MF and even in tumor-stage MF.

2.3. Light sources

There is a wide spectrum of light sources for PDT, including continuous-wavelength light sources (red, blue, white, or green), lasers, incoherent light sources and so on. To obtain desired therapeutic effects, the photosensitizing porphyrins should be activated by light of an appropriate wavelength. The porphyrins or related photosensitizers exhibit a very typical absorption spectrum with the highest peak at approximately 405 nm in the blue wavelength region, the so-called Soret-band, and several Q-bands with an absorption peak at 635 nm in the red wavelength region [19]. It has been reported that compared with the blue spectrum, the red light shows better tissue penetration [20].

Moreover, laser and incoherent light sources can be used in PDT. The advantage of laser is shorter exposure time compared with incoherent light sources, though the laser cannot ideally match the porphyrin absorption spectrum. The development of intense pulsed light (IPL) devices (e.g. PDT 1200l, Waldmann Medizintechnik) enables application of incoherent light sources for PDT, with an emission spectrum ranging from 500 to 1300 nm. At present, the preferred light in topical PDT is incoherent light sources with wide illumination fields because this can accomplish the simultaneous irradiation of larger areas. Several studies have reported successful treatments of mycosis fungoides with IPL device for PDT [21,22]. In addition, the LED system of incoherent light sources (e.g. Aktilite™, Galderma, Alby- sur-Che íran, France; Omnilux PDT™, Phototherapeutics, Altrincham, UK) is a promising choice because it emits only a narrow band of wavelengths that perfectly matches the absorption spectrum of porphyrins.

In addition, there is another type of photosensitizer for mycosis fungoides treatment under investigation. The silicon phthalocyanine Pc4, a second-generation photosensitizer, allows deeper tissue penetration of red light ($\lambda = 675$ nm; $\varepsilon = 2 \times 10^5$) when compared to PpIX or photofrin whose peak absorption in the red region is at 630–635 nm [23]. Pc4 accumulates in mitochondria to damage the anti-apoptotic proteins Bcl-2 and Bcl-xL, releases cytochrome c into the cytosol, and induces cellular apoptosis [24]. A phase 1 clinical trial of Pc4-PDT for mycosis fungoides was performed in 2010 and

Table 1
Studies demonstrating the effect of PDT in the treatment of relapsed or refractory MF.

Study	Number of lesions/Number of patients	Previous treatments	Light/Light doses	Photosensitizer/ Application duration	Number of PDT sessions	Type of lesions	Percentage of CR	Follow-up period (months)
Wolf et al. [26]	3/2	Topical steroids, UVB phototherapy, and PUVA	Visible light/40 J/cm^2	20% ALA cream/4–6 h	4–5	3 plaques	CR 3/3 (100%)	3–6
Wang et al., [27]	3/1	Nine cycles of chemotherapy	Laser, 635 nm/60 J/cm^2	20% ALA cream/4–6 h	3	3 plaques	CR 3/3 (100%)	33
Orenstein et al. [28]	5/1	Nitrogen mustard and electron beam radiation	Incoherent light, 580–720 nm/380J/cm^2	20% ALA cream/16 h	1	5 plaques	CR 5/5 (100%)	24
Markham et al. [21]	1/1	Oral 8-methoxypsoralen (8-MOP) plus PUVA	Incoherent light, 580–740 nm/20J/cm^2	20% ALA cream/4 h	5	1 tumor	CR 1/1 (100%)	12
Leman et al.[29]	2/1	Clobetasol propionate twice daily for 6 weeks	Red light, 630± 15 nm/100J/cm^2	20% 5-ALA cream/24 h	4	2 patches	CR 2/2 (100%)	12
Coors et al.[22]	7/4	PUVA, interferon alfa-2a, and acitretin	Incoherent light/72-144J/cm^2	20% 5-ALA cream/6 h	1–7	5 plaques, 2 tumors	CR 7/7 (100%)	14–18
Zane et al. [30]	5/5	Topical steroids, localized PUVA and UVA1 phototherapy	Red light, 635± 18 nm/37.5J/cm^2	20% MAL cream/3 h	1–9	5 patches	CR 4/5 (80%) PR 1/5	12–34
Edström et al. [31]	10/8	PUVA, local steroids, and radiotherapy	Incoherent light, 600–730 nm/80-180J/cm^2	20% 5-ALA cream/5–6 h	2–11	8 plaques, 2 tumors	6/8 plaques CR (75%), 0/2 tumors CR (0%)	87
Recio et al. [32]	2/2	Topical steroids, PUVA, and imiquimod cream	Laser, 585 nm/8 J/cm^2	20% ALA cream/4 h	3	2 plaques	CR 2/2 (100%)	24
Hegyi et al. [33]	1/1	Photochemotherapy and local therapy with corticosteroids	Red light, 635 nm/100J/cm^2	20% MAL cream/3 h	3	1 plaque	CR 1/1 (100%)	16
Debu et al. [34]	1/1	Topical steroids, PUVA, retinoids plus PUVA, methotrexate, bexarotene and rexinoid	Red light, 635 nm/37J/cm^2	20% MAL cream/3 h	4	1 plaque	PR 1/1 (100%)	19 (died of unrelated causes)
Fernández-Guarino et al. [35]	24/12	Topical corticosteroids, topical immunomodulators, topical retinoid, and PUVA	Red light, 630 nm/37J/cm^2	16% MAL cream/3 h	5–6	24 plaques	CR 12/24 (50%)	6–22
Kim et al. [36]	11/7	Topical steroids, PUVA, ultraviolet A1 (UVA1), and oral retinoids	Red light, 630 nm/37.5 J/cm^2	16.8% MAL cream/3 h	2–6	9 plaques, 2 patches	3/9 plaques CR (33.3%) 1/2 patches CR (50%)	8–31
Debu et al. [37]	5/3	Resistant to previous treatments	Red light, 630 nm/37 J/cm^2	16.8% MAL cream/3 h	1–7	4 plaques, 1 patches	3/4 plaques CR (75%), 1/1 patch CR (100%)	12–28
Calzavara-Pinton et al. [38]	19/19	Resistant to previous treatments	Red light, 635± 18 nm/37 J/cm^2	20% MAL cream/3–4 h	1–7	19 plaques	CR 5/19 (26%)	20 (2 CR patients relapsed)
Ariel et al. [39]	1/1	Psoralen + ultraviolet A, bexarotene, and total-skin electron beam radiation	Incoherent light, 570–670 nm/37J/cm^2	160 mg/g MAL cream/3 h	3	1 tumor	CR 1/1 (100%)	60
Quéreux et al. [40]	12/7	Bexarotene, and interferon	Red light, 630 nm/37 J/cm^2	20% MAL cream/3 h	2–6	5 plaques, 7 patches	5/5 plaques CR (100%), 4/7 patches CR (57.1%)	10–35
Jung et al. [41]	1/1	Eight treatment sessions with a 308-nm excimer laser	Red light, 630 nm/37 J/cm^2	20% MAL cream/1.5 h	8	1 tumor	CR 1/1 (100%)	/
Jang et al. [42]	1/1	Topical steroid	Red light, 570–670 nm/37 J/cm^2	MAL cream/4 h	2	1 patch	CR 1/1 (100%)	/
Han et al. [43]	3/3	PUVA, and interferon	Red light, 635 nm/60nw/cm^2	20% ALA solution/4 h	2–3	3 plaques	CR 2/3 (66.7%)	8–17

Abbreviations: ALA, aminolevulinic acid; MAL, methyl-aminolevulinic acid; PUVA, psoralen-UV-A; CR, complete response; PR, partial response.

demonstrated the potential usefulness of topically delivered Pc4 to treat MF with good tolerability and promising efficacy [25].

3. Result

We found and reviewed 20 published articles to show that PDT was useful as a skin-limited treatment in localized forms of MF that responded poorly to regular treatments. Reported cases of successful treatment of relapsed or refractory MF plaques using PDT are summarized in Table 1. These studies included a total of 81 patients previously treated with topical steroids, photochemotherapy, nitrogen mustard and so on. These patients had 117 relapsed or refractory MF lesions which were treated with PDT, 78.6% (92/117) of which were plaque-stage MF, 15.4%(18/117) of which were patch-stage MF, and the remaining 6%(7/117) of which were tumor-stage MF.

Since the first case was reported in 1994 [26] and through 2006, 20% 5-ALA was used as the photosensitizer for PDT with 4–16 h occlusion time. There were 9 studies using ALA, and 11 studies using MAL. 5-ALA has been substituted by MAL [30] with 3 h occlusion time. MAL has increased lipophilicity and resulted in deeper skin penetration. The majority of these studies used long wavelengths around 600 nm by various light sources (red light, laser, and incoherent light), which could penetrate deeper and be more effective for the treatment of thicker lesions. The frequency of the treatment sessions was around every 2 to 4 weeks, and almost all the studies described the need for multiple sessions. Number of sessions was associated with total light doses, type of lesions, and sensitivity to the treatment. Repeated sessions of PDT seemed to be mandatory in treating MF.

The overall complete response (CR) rate of the 117 MF lesions was 63.2% (60.9% in plaques, 72.2% in patches, and 71.4% in tumors, respectively). In theory, the correlation between CR rate and infiltrate depths of lesions should be negative. However, the publication bias could account for these observations. Edström et al. observed a lower rate of response in tumor stage lesions, though all the tumor lesions showed regression [31]. Follow-up period of each study varies from 3 months post-treatment to 87 months. Therefore, PDT can be a valuable treatment option for relapsed or refractory MF at different clinical stages, especially for patch or plaque-stage, although optimal parameters of PDT have not yet been defined.

4. Discussion

For mycosis fungoides, multiple treatment modalities such as PUVA, interferon, systemic chemotherapy, and total body electron beam irradiation may fail to control all lesions in some patients, so additional localized therapies may be required. PDT is a recently introduced therapeutic modality that has been proven effective and well tolerated in patients with localized relapsed or refractory MF lesions, even in tumor-stage. The advantages of PDT include excellent cosmetic results, noninvasive nature, excellent selectivity, low risk of toxicity accumulation, and negligible generalized photosensitivity, and low carcinogenic potential. These advantages make PDT a valuable therapeutic option in difficult-to-treat subsets of MF, such as relapsed or refractory MF, or even in sensitive areas such as the face and neck [34].

In conclusion, PDT is a promising treatment option for relapsed or refractory mycosis fungoides. Despite the relative simplicity of the technique, established treatment protocols have not yet been optimized for PDT in MF with respect to ALA or MAL application time, light dosimetry, delivery, frequency of treatment, and number of sessions. Therefore, larger clinical studies are required and the optimal treatment regimens need to be clarified.

Funding support

There is no funding support for this review article.

Conflict of interest disclosure

The authors do not have conflicts of interest to disclose.

Acknowledgements

There is no funding support for this review article.

References

[1] M. Girardi, P.W. Heald, L.D. Wilson, The pathogenesis of mycosis fungoides, N. Engl. J. Med. 350 (2004) 1978–1988.
[2] P.T. Barton, S.S. Devesa, W.F. Anderson, J.R. Toro, Cutaneous lymphoma incidence patterns in the United States: a population based study of 3884 cases, Blood 113 (2009) 5064–5073.
[3] M.A. Weinstock, J.W. Horm, Mycosis fungoides in the United States, JAMA 60 (1988) 42–46.
[4] M.M. Morales Suarez-Varela, A. Llopis Gonzalez, A. Marquina Vila, J. Bell, Mycosis fungoides: review of epidemiological observations, Dermatology 201 (2000) 21–28.
[5] S.M. Horwitz, E.A. Olsen, M. Duvic, P. Porcu, Y.H. Kim, Review of the treatment of mycosis fungoides and sézary syndrome: a stage-based approach, J. Natl. Compr. Canc. Netw. 6 (April) (2008) 436–442.
[6] N.S. Agar, E. Wedgeworth, S. Crichton, et al., Survival outcomes and prognostic factors in mycosis fungoides/Sézary syndrome: validation of the revised International Society for Cutaneous Lymphomas/European Organisation for Research and Treatment of Cancer staging proposal, J. Clin. Oncol. 28 (2010) 4730–4739.
[7] S.H. Swerdlow, E. Campo, N.L. Harris, et al., WHO classification of tumours of haematopoietic and lymphoid tissues, 4th ed., IARC Press, Lyon, 2008.
[8] W. Kempf, C.A. Sander, Classification of cutaneous lymphomas – an update, Histopathology 56 (2010) 57–70.
[9] K. Svanberg, T. Andersson, D. Killander, et al., Photodynamic therapy of non-melanoma malignant tumours of the skin using topical delta-amino levulinic acid sensitization and laser irradiation, Br. J. Dermatol. 130 (1994) 743–751.
[10] W.H. Boehncke, K. Konig, A. Ruck, R. Kauffman, W. Sterry, In vitro and in vivo effects of photodynamic therapy in cutaneous T cell lymphoma, Acta Derm. Venereol. 74 (1994) 201–205.
[11] K. Rittenhouse-Diakun, H. Van Leengoed, J. Morgan, et al., The role of transferrin receptor (CD71) in photodynamic therapy of activated and malignant lymphocytes using the heme precursor delta-amino-levulinic acid (ALA), Photochem. Photobiol. 61 (1995) 523–528.
[12] F. Gad, G. Viau, M. Boushira, R. Bertrand, R. Bissonnette, Photodynamic therapy with 5-aminolevulinic acid induces apoptosis and caspase activation in malignant T cells, J. Cutan. Med. Surg. 5 (2001) 8–13.
[13] A. Orenstein, J. Haik, J. Tamir, et al., Photodynamic therapy of cutaneous lymphoma using 5-aminolevulinic acid topical application, Dermatol. Surg. 26 (2000) 765–769.
[14] J.C. Kennedy, R.H. Pottier, D.C. Pross, Photodynamic therapy with endogenous protoporphyrin IX: basic principles and present clinical experience, J. Photochem. Photobiol. B 6 (1990) 143–148.
[15] P. Babilas, S. Karrer, A. Sidoroff, M. Landthaler, R.M. Szeimies, Photodynamic therapy in dermatology–an update, Photodermatol. Photoimmunol. Photomed. 21 (2005) 142–149.
[16] P. Babilas, S. Schreml, M. Landthaler, R.M. Szeimies, Photodynamic therapy in dermatology: state-of-the-art, Photodermatol. Photoimmunol. Photomed. 26 (2010) 118–132.
[17] Q. Peng, A.M. Soler, T. Warloe, J.M. Nesland, K.E. Giercksky, Selective distribution of porphyrins in skin thick basal cell carcinoma after topical application of methyl 5-aminolevuli- nate, J. Photochem. Photobiol. B 62 (2001) 140–145.
[18] Ariel Hasson, Cristián Navarrete-Dechent, Claudia Nicklas, et al., Topical photodynamic therapy with 5-aminolevulinate for the treatment of tumor-stage mycosis fungoides: a case report, Int. J. Dermatol. 52 (2013) 1535–1537.
[19] R.M. Szeimies, C. Abels, C. Fritsch, et al., Wavelength dependency of photodynamic effects after sensitization with 5-aminolevulinic acid in vitro and in vivo, J. Invest. Dermatol. 105 (1995) 672–677.
[20] S.B. Brown, The role of light in the treatment of non-melanoma skin cancer using methyl aminolevulinate, J. Dermatolog. Treat. 14 (Suppl. 3) (2003) 11–14.
[21] T. Markham, K. Sheahan, Collins P. Topical 5-aminolaevulinic acid photodynamic therapy for tumour-stage mycosis fungoides, Br. J. Dermatol. 144 (2001) 1262–1263.

[22] E.A. Coors, P. on den Driesch, Topical photodynamic therapy for patients with therapy-resistant lesions of cutaneous T-cell lymphoma, J. Am. Acad. Dermatol. 50 (2004) 363–367.
[23] N.L. Oleinick, A.R. Antunez, M.E. Clay, B.D. Rihter, M.E. Kenney, New phthalocyanine photosensitizers for photodynamic therapy, Photochem. Photobiol. 57 (1993) 242–247.
[24] N.L. Oleinick, R.L. Morris, I. Belichenko, The role of apoptosis in response to photodynamic therapy: what, where, why and how, Photochem. Photobiol. Sci. 1 (2002) 1–21.
[25] E.D. Baron, C.L. Malbasa, D. Santo-Domingo, et al., Silicon phthalocyanine (Pc 4) photodynamic therapy is a safe modality for cutaneous neoplasms: results of a phase 1 clinical trial, Lasers Surg. Med. 42 (2010) 728–735.
[26] P. Wolf, R. Fink-Puches, L. Cerroni, H. Kerl, Photodynamic therapy for mycosis fungoides after photosensitization with 5-aminolevulinic acid, J. Am. Acad. Dermatol. 31 (1994) 678–680.
[27] I. Wang, B. Bauer, S. Anderson-Engels, S. Svanberg, K. Svanberg, Photodynamic therapy utilizing delta aminolevulinic acid in non-melanoma skins malignancies of the eyelid and periocular skin, Acta Ophthalmol. Scand. 77 (1999) 182–188.
[28] A. Orenstein, J. Haik, J. Tamir, et al., Photodynamic therapy of cutaneous lymphoma using 5-aminolevulinic acid topical application, Dermatol. Surg. 26 (2000) 765–769.
[29] J.A. Leman, D.C. Dick, C.A. Dick, Topical 5-ALA therapy for the treatment of cutaneous T-cell lymphoma, Clin. Exp. Dermatol. 27 (2002) 516–518.
[30] C. Zane, M. Venturini, R. Sala, P. Calzavara-Pinton, Photodynamic therapy with mehtylaminolevulinate as a value treatment option for unilesional cutaneous T-cell lymphoma, Photodermatol. Photoinmunol. Photomed. 22 (2006) 254–258.
[31] D.W. Edström, M.A. Hedblad, Long-term follow-up of photodynamic therapy for mycosis fungoides, Acta Derm. Venereol. 88 (2008) 288–290.
[32] E.D. Recio, B. Zambrano, M.L. Alonso, et al., Topical 5-aminolevulinic acid photodynamic therapy for the treatment of unilesional mycosis fungoides: a report of two cases and review of the literature, Int. J. Dermatol. 47 (2008) 410–413.
[33] J. Hegyi, T. Frey, P. Arenberger, Unilesional mycosis fungoides treated with photodynamic therapy: a case report, Acta Dermatovenerol. Alp. Panonica Adriat. 17 (2008) 75–78.
[34] A. Debu, C. Girard, N. Kluger, B. Guillot, O. Dereure, Topical methyl aminolaevulinate-photodynamic therapy in erosive facial mycosis fungoides, Br. J. Dermatol. 163 (2010) 884–885.
[35] M. Fernández-Guarino, A. Harto, B. Pérez-García, C. Montull, E. De Las Heras, P. Jaén, Plaque-phase mycosis fungoides treated with photodynamic therapy: results from 12 patients, Actas Dermosifiliogr. 101 (2010) 785–791.
[36] S.T. Kim, D.Y. Kang, J.S. Kang, J.W. Baek, Y.S. Jeon, K.S. Suh, Photodynamic therapy with methyl-aminolaevulinic acid for mycosis fungoides, Acta Derm. Venereol. 92 (2012) 264–268.
[37] A. Debu, D. Bessis, C. Girard, A. Du Thanh, B. Guillot, O. Dereure, Photodynamic therapy with methyl aminolaevulinate for cervical and/or facial lesions of folliculotropic mycosis fungoides: interest and limits, Br. J. Dermatol. 168 (2013) 896–898.
[38] P.G. Calzavara-Pinton, M.T. Rossi, R. Sala, Italian Group For Photodynamic Therapy, A retrospective analysis of real-life practice of off-label photodynamic therapy using methyl aminolevulinate (MAL-PDT) in 20 Italian dermatology departments. Part 2: oncologic and infectious indications, Photochem. Photobiol. Sci. 12 (2013) 158–165.
[39] Ariel Hasson, Cristián Navarrete-Dechent, Claudia Nicklas, et al., Topical photodynamic therapy with 5-aminolevulinate for the treatment of tumor-stage mycosis fungoides: a case report, Int. J. Dermatol. 52 (2013) 1535–1537.
[40] G. Quéreux, A. Brocard, M. Saint-Jean, et al., Photodynamic therapy with methyl-aminolevulinic acid for paucilesional mycosis fungoides: a prospective open study and review of the literature, J. Am. Acad. Dermatol. 69 (2013) 890–897.
[41] S.E. Jung, Y.C. Kim, A case of pagetoid reticulosis treated with fractional laser-assisted photodynamic therapy, Clin. Exp. Dermatol. 40 (2015) 468–470.
[42] S. Jang, E.J. Hwang, H.H. Ryu, K.H. Cho, S.J. Jo, A case of unilesional mycosis fungoides treated with photodynamic therapy using methyl-Aminolevulinate, Korean J. Dermatol. 53 (2015) 58–61.
[43] D. Han, J. Xue, T. Wang, Y. Liu, Observation of clinical efficacy of photodynamic therapy in 3 patients with refractory plaque-stage mycosis fungoides, Photodiagn. Photodyn. Ther. 16 (2016) 9–11.